Orwell Bay

Shoreline length: 39 km

Cliffs and wharf remnants on north shore of Point Prim

Cliff at Belfast Highlands Greens Golf (left) and Lord Selkirk Park (right)

Cliff to wetland transition towards inner Orwell Bay

Shore Types

Orwell Bay is the north facing portion of the Point Prim peninsula located on the southeast end of PEI, with a majority of wetlands and cliffs.

% Shore types within SMU

low plain	bluff	sloped armour	cliff	wetland	vertical struct.	dunes	sandspit
12	4	0	33	50	0	1	0

Governance, Infrastructure, and Economy

This SMU is almost completely located within the Rural Municipality of Belfast (92%). Key infrastructure includes a portion of the Trans-Canada Highway and provincial roads: Route 209 (Point Prim Road, at Eldon and Prim Point). Coastal economic drivers include the Belfast Highland Greens golf course. Local and legacy land use constraints include a few small, developed lots, several undeveloped small lots, and unpaved coastal roads.

Natural Assets and Heritage

The Lord Selkirk Provincial Park is within Orwell Bay. The shoreline includes critical habitat for bird species including the threatened Bank Swallow.

Planning and Policy Framework

Provincial rules apply. Land use and development is managed under provincial development standards, which include horizontal coastal building setbacks based on edge of feature and a calculation of erosion over time, building setbacks for dunes, and a requirement for a coastal buffer for new subdivisions.

There are no specific flood elevations identified in regulations, but coastal hazard assessments are typically conducted.

Coastal Flood and Erosion Risk

Infrastructure near the wetlands within in this area are vulnerable to increasing flood hazards. Erosion hazard is limited due to the limited amount of assets and sheltered orientation of the area up the bay. The following flood risk indicator is based on the even with 1% annual exceedance probability (100-year return) with increasing amounts of sea level rise (SLR). The erosion risk indicator is based on potential shoreline change or salt marsh migration to future time horizons.

Number of at-risk buildings [N. per km shoreline]

	Present	0.3m SLR (2050)	0.6m SLR (2080)	1.2m SLR (2130)
SMU Orwell Bay				
Flood risk	0	0	0	4 [<1/km]
Erosion risk	NA	5 [<1/km]	5 [<1/km]	6 [<1/km]

Sediment transport - The estimated net sediment transport rate averaged across the SMU is moderate. Sediment transport rate potential is highest along the north facing peninsula.

Proposed SMP Strategies

The primary strategy for Orwell Bay is No Active Intervention, followed by long-term Managed Realignment for the limited number of assets at risk.

	Short-term	Medium-term	Long-term
Sea level rise range	0 – 0.3 m	0.3 – 0.6 m	0.6 – 1.2 m
Potential timeframe	2030 - 2050	2050 - 2080	2080 - 2130
Proposed SMU strategy	**NAI**	**MR**	**MR**
Rationale	Mid- to long-term MR will reduce risk to the limited number of vulnerable assets, allowing salt marsh migration up the estuary.		

LEGEND

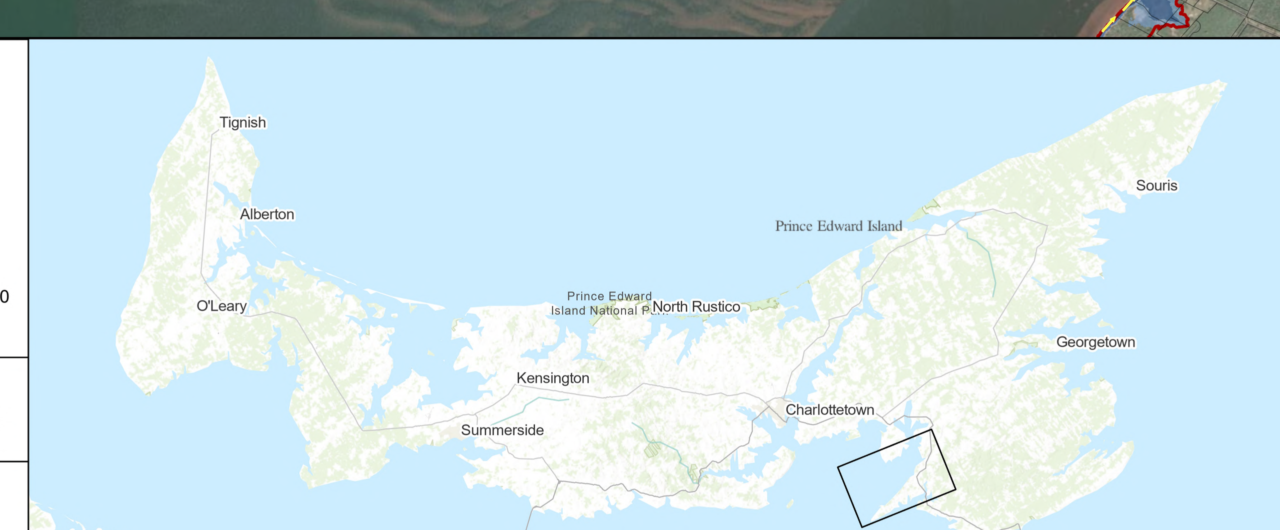
- Shoreline Management Unit
- Net Longshore Sediment Transport
- High Hazard Area
- Flood Extent - Present Day (100-year Return)
- Flood Extent - 1.2m SLR (100-year Return)
- Property Boundary
- Building > 90ft²

N W E S

0 260 520 1,040
Metres

Coordinate System:
NAD 83 CSRS PEI Double Stereographic

NOTES:



COASTAL EROSION AND FLOOD HAZARD

Shoreline Management Unit: Orwell Bay

Prince Edward Island
Île-du-Prince-Édouard

CBCL